CAPTCHA Inspector - Capabilities, Requirement Coverage and Roadmap

Technology Brainz assignment · status of the current build

1. What the tool does today

Capability	Current behaviour
Import the CSV	Reads the 200-row CSV, validates it, keeps every column, reports bad rows with their row number, normalises the expected CAPTCHA label.
Open Contact Us pages	Real headless Chromium (Playwright); waits for the page to settle; reads the rendered page, scripts and iframes.
Identify CAPTCHAs	Google reCAPTCHA (v2 / invisible), hCaptcha, Cloudflare Turnstile, and other anti-bot challenges; records the type and widget state; ignores mentions that appear only in static code.
Keep the workflow moving	A CAPTCHA never blocks the batch: it is recorded with a screenshot, queued in Human Review, and the scan continues with the next site.
Compare with the CSV	Expected vs Detected: Match, Mismatch, Not detected, Not accessible, Not verified. Percentages use only inspected rows and state the sample size.
Inspect forms safely	Counts forms and fields; never clicks CAPTCHAs, never submits forms, never collects tokens.
Solve text CAPTCHAs	OCR pipeline for our own synthetic text CAPTCHAs (about 99.3% on our own generator).
Verify tokens	Calls the providers' official verification service with their published test keys and records the real answer.
Reports	Audit report and Expected-vs-Detected report (CSV / Markdown), safe against spreadsheet formula injection.
Safety and audit	Domain allow-list, SSRF protection, robots.txt respected, request pacing per host, role-based login, tamper-evident audit trail.

2. Requirement coverage

Requirement (from the assignment message)	Status	Notes
Import the 200-row CSV	**Met**	Tested with the assignment's column layout.
Use Playwright	**Met**	All page inspection runs through Playwright Chromium.
Work with Contact Us forms	**Met**	Uses the CSV's Contact Page value; discovers contact pages from a home page when asked.
Identify the CAPTCHA challenge	**Met**	Provider, type and state; checked on fixtures and on real sites.
CAPTCHA must not prevent the workflow from proceeding	**Met**	Recorded, queued for a person, batch continues. Unreachable sites are reported, not fatal.

Handle different CAPTCHA types	**Partly**	All major types are **identified and handled by routing to a human**. Automated solving exists only for our own text CAPTCHAs.
Solve third-party CAPTCHAs (reCAPTCHA, hCaptcha, Turnstile)	**Not done, by design**	Bypassing them would breach the providers' terms and the websites' access controls. Not implemented.
Work on all 200 websites	**Partly**	Every row gets a result. Sites that block automated browsers, forbid crawling in robots.txt, or time out are reported as "Not accessible" with the reason. A full 200-site run has not been completed yet; the live sample so far is small (4 sites: 2 inspected and matching, 2 not accessible).
Real-time use	**Partly**	Works locally and can be hosted; a live deployment and allow-list for the 200 domains still have to be set up.

Overall. The detection, evidence, reporting and non-blocking workflow parts are complete and tested (backend suite: 630 passed, 1 skipped, 0 failed; browser suites 30/30, 32/32, 19/19, 16/16 at the last full run). The one requirement that is not met is automated solving of third-party CAPTCHAs, and the share of the 200 sites that can be inspected depends on how many allow automated access.

3. Known limits

Limit	Why
Some sites show "Not accessible"	Bot walls, 403/429 responses, robots.txt disallow, timeouts. The tool stops rather than evading them.
No automatic solving of reCAPTCHA / hCaptcha / Turnstile	Legal and terms-of-service reasons; a person reviews each one.
Confidence per page is not shown	It is not measured, so the screens show "-" instead of an invented number.
Small live sample	The real-site results so far are not a statistical accuracy claim.
Local-only until hosted	No public link exists yet.

4. Possible future additions

Feature	Value
Run and publish the full 200-site scan	Real coverage numbers (inspected / not accessible / matched) for the employer.
Hosted deployment (Docker, PostgreSQL)	Live link for reviewers; allow-list preconfigured for the CSV domains.
Retry and resume for batches	Retry transient failures and continue an interrupted 200-site run.
Permission-based CAPTCHA handling	For sites the client owns or has written permission to test: use the providers' test or allow-list mechanisms, or a legally reviewed, provider-sanctioned integration. Needs client sign-off first.
Reviewer workflow tools	Assign review items, SLA timers, bulk approve/reject, comments.

Notifications	Email or chat alert when a batch finishes or a new CAPTCHA type appears.
Scheduled re-scans and trends	Detect when a site adds, removes or changes its CAPTCHA over time.
Larger accuracy benchmark	Labelled sample of real pages to measure detection precision and recall.
Export to PDF/Excel dashboards	One-click executive report from the Reports screen.
Stronger accounts	Password policy, per-user API keys, single sign-on, expiry for reviewer accounts.